

RAINA DAS

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PROFESSIONAL SUMMARY

AI/ML enthusiast with a strong foundation in Electronics and Communication Engineering (ECE). Driven to integrate machine learning with electronics, applying algorithms in small projects and exploring innovative solutions. Focused on advancing expertise to address real-world challenges through the intersection of ML and electronics.

EDUCATION

B.TECH (Electronics and Communication) | **Institute Of Engineering & Management, Kolkata**

2023-2027

CGPA: 9.27/10 (till 5th semester)

ISC | **ST Paul's Boarding & Day School, Kolkata** | 2023 | 91%

ICSE | **Vivekananda Mission School** | 2021 | 93.6%

TECHNICAL SKILLS

Programming Languages: Python, C, Java

HDL: Verilog (RTL design – beginner)

VLSI / EDA Tools: Cadence, Synopsys (academic exposure)

Backend: FASTAPI

Databases: SQL, ChromaDB

Library: Pandas, Numpy, Scikit-Learn, Matplotlib

AI Tools: LangChain

PERSONAL PROJECTS

SPAM EMAIL FILTERING

- **Problem Statement:** Engineered a machine learning system to replace legacy rule-based email filters, addressing the security risks of evolving phishing and spam text patterns.
- **Tools & Technologies:** Python, Scikit-learn, Naive Bayes, NLTK, TF-IDF Vectorization, Pandas.
- **Results:** Built an end-to-end NLP pipeline achieving **98.2% accuracy**

AI-BASED IPC SECTION RECOMMENDATION SYSTEM FOR LEGAL QUERIES

- Developed NLP-based legal query recommendation system mapping natural language inputs to IPC sections, leveraging Python, FastAPI, SQL, and Chroma for semantic search and preprocessing pipelines
- Improved retrieval accuracy through vector similarity and contextual matching techniques, enabling precise and relevant legal information recommendations

PERSONALIZED AI TUTORING

- Developed a GenAI-powered personalized tutoring assistant using LangChain and the Mistral API.
- Implemented semantic memory and context-based response generation using ChromaDB vector storage.
- Built retrieval-based conversational workflows for adaptive learning and contextual question answering.
- Automated Assessments-students can track their progress by attempting the quizzes generated based on user topic history.
- Integrated backend APIs for efficient prompt handling and response orchestration.

CERTIFICATIONS

- NPTEL Online Certification -Machine Learning(IIT MADRAS)
- IBM Skillsbuild Foundation Virtual Internship Certification On Artificial Intelligence(AI)
- Winter Internship – Microelectronics & VLSI Design
Jadavpur University | December 2025-January 2026
 - Learning VLSI design flow, semiconductor devices, and fabrication basics
 - Gaining exposure to Verilog-based design and industry-standard EDA tools (Cadence, Synopsys)